

# TinyStories TiDAR Greedy Acceptance Comparison

## Run Summary

All 4 runs branch from the stable run at step 90,000:

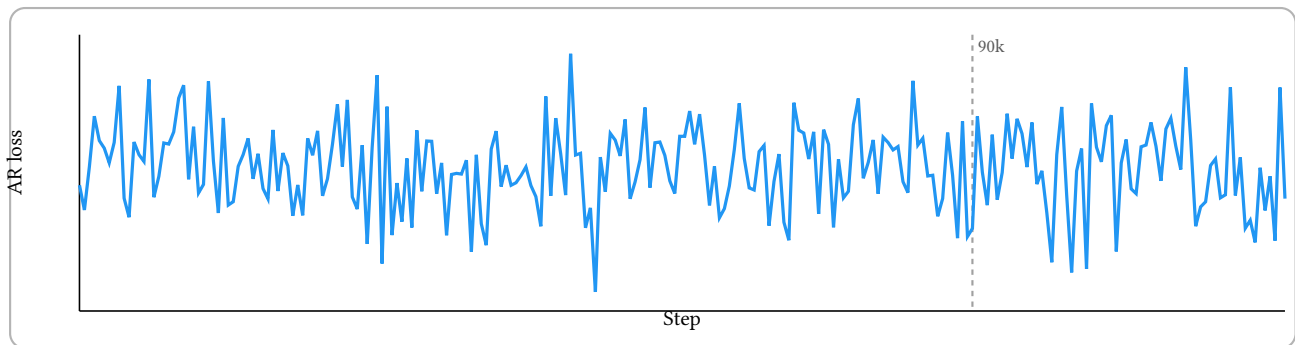
- **Stable (blue)**:  $\alpha=1$ ,  $\beta=1$  – baseline AR+Diff training, continued to 121k
- **KL-only (red)**:  $\alpha=0.2$ ,  $\beta=0$ ,  $\rho=0.5$ ,  $\delta=0.3$  – forward KL + greedy, no diff loss
- **KL-keep (green)**:  $\alpha=1$ ,  $\beta=1$ ,  $\rho=0.1$ ,  $\delta=0.3$  – forward KL + greedy, keeps AR+Diff
- **Distill (purple)**:  $\alpha=1$ ,  $\beta=1$ ,  $\eta=0.04$ ,  $T=2$  – soft distillation AR→Diff

## Legend

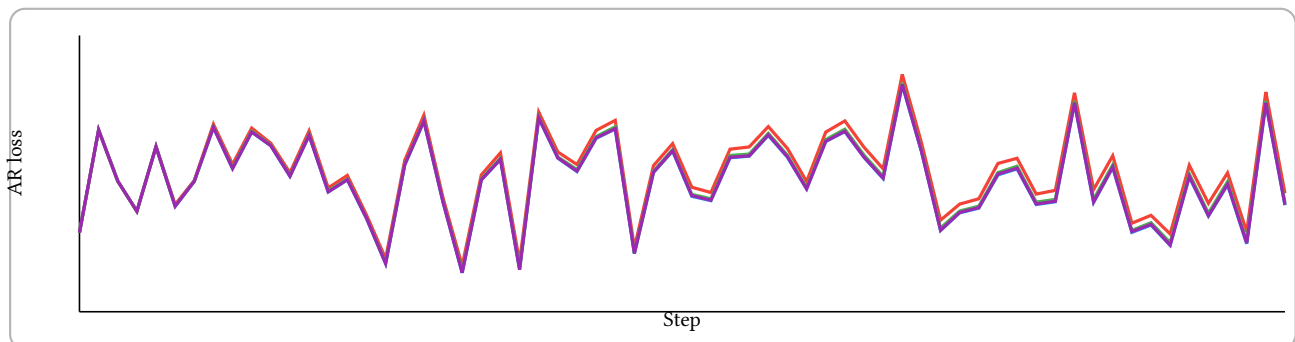
- |                                                                     |                                                                                   |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <span style="color: blue;">■</span> Stable (1, 1, 0, 0, 0, 0)       | <span style="color: green;">■</span> KL-keep (1, 1, 0.1, 0, 0.3, 0)               |
| <span style="color: red;">■</span> KL-only (0.2, 0, 0.5, 0, 0.3, 0) | <span style="color: purple;">■</span> Distill (1, 1, 0, 0, 0, $\eta=0.04$ $T=2$ ) |

## AR loss (raw)

Stable 0-121k

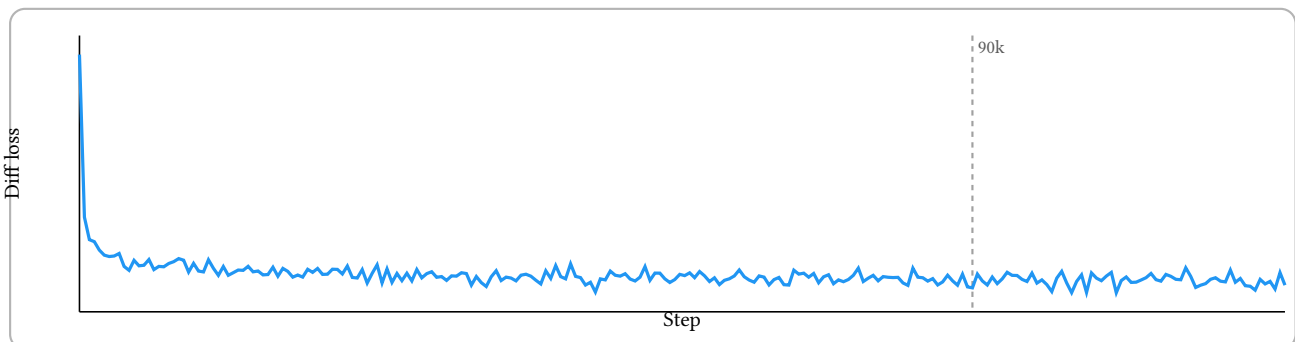


All runs 90k-121k

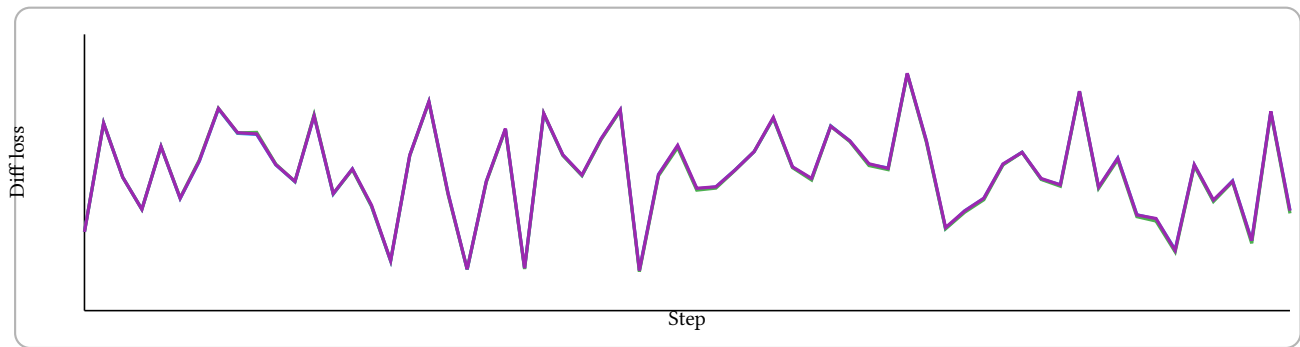


## Diff loss (raw)

Stable 0-121k



All runs 90k-121k

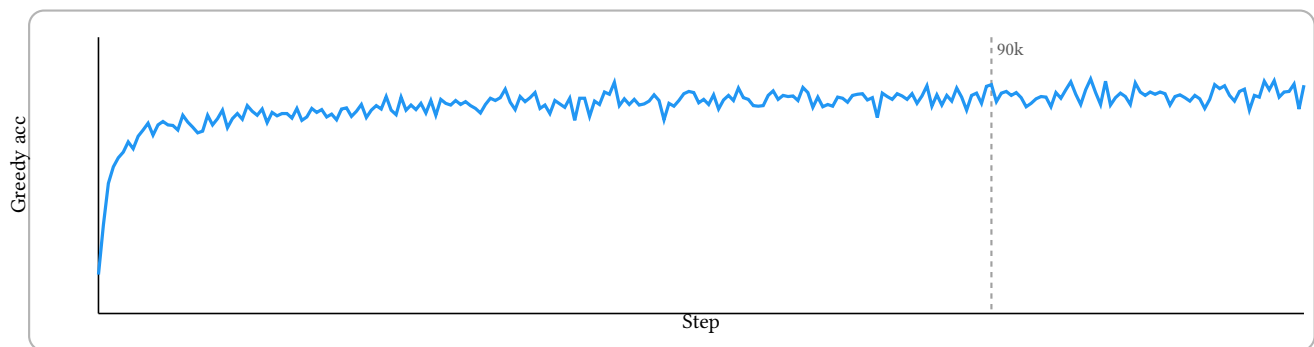


KL-only diff=0 after 90k, excluded from the zoomed panel.

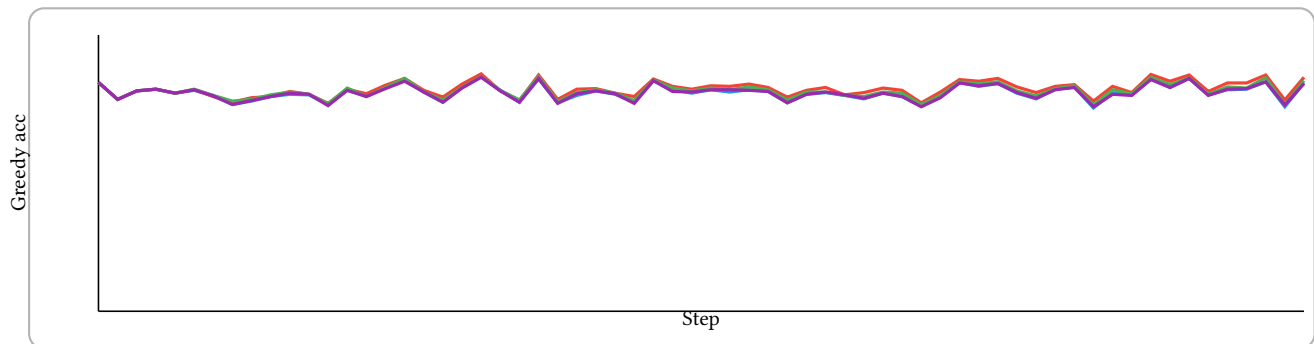
## Greedy acceptance

Greedy acceptance = fraction of positions where  $\text{argmax}(\text{AR}) == \text{argmax}(\text{Diff})$ .

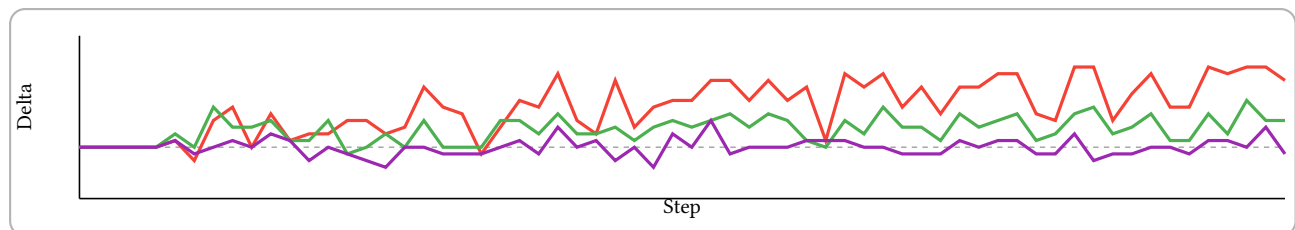
Stable 0-121k



All runs 90k-121k



Delta vs stable (90k-121k)



## Inference Results

Greedy decoding (temperature=0) with draft\_len=6.

Prompts (10):

- P0: “There was a small village”
- P1: “Once upon a time, a little girl”
- P2: “The boy was very happy because”
- P3: “In the forest, a tiny fox”
- P4: “The cat and the dog were”
- P5: “One day, the teacher said”
- P6: “After school, the kids went”
- P7: “The robot wanted to learn”
- P8: “On a sunny morning, Emma”
- P9: “The brave knight looked at”

**Accept/Iter summary (mean / best / worst across prompts)**

| Checkpoint    | Steps=50           | Steps=100          | Steps=300          |
|---------------|--------------------|--------------------|--------------------|
| Stable @90k   | 2.41 / 3.50 / 1.63 | 2.26 / 2.75 / 1.71 | 2.10 / 2.45 / 1.88 |
| Stable @121k  | 2.22 / 3.27 / 1.53 | 2.09 / 2.68 / 1.60 | 2.10 / 2.69 / 1.57 |
| KL-only @121k | 2.42 / 3.27 / 1.81 | 2.30 / 2.68 / 1.80 | 2.08 / 2.49 / 1.61 |
| KL-keep @121k | 2.40 / 3.27 / 1.75 | 2.29 / 2.83 / 1.80 | 2.18 / 2.90 / 1.75 |
| Distill @121k | 2.31 / 3.50 / 1.53 | 2.26 / 3.09 / 1.87 | 2.19 / 2.74 / 1.80 |